SG98 Remote / Controller Connection Matrix

Bench run date: 2026-04-26 17:43 PDT

Primary goal: no factory-fresh remote/controller pair should strand a headless controller.

Area	Observed / Build	Meaning
Live remote	Waveshare AMOLED 1.75 / /dev/cu.usbmodem21101	Factory/no-info reset before each pass: no saved Wi-Fi and no locked ESP-NOW peer.
Full-size pass	Adafruit Feather Controller / controller_box	Same physical controller with screen, buttons, and encoder enabled. OFF remains allowed because local recovery exists.
Headless-sim pass	controller_box_headless_sim	Same physical controller with board features reporting no screen, no buttons, no encoder, no encoder button.
Steady state	ESP-NOW channel 1	Both passes ended with remote online over ESP-NOW, remote Wi-Fi off, and no saved home Wi-Fi credentials.

Build Target	Result	Notes
remote-waveshare-175	PASS	Built and flashed to the bench remote.
controller-universal-s3	PASS	Built and flashed for the live full-size controller pass.
controller-universal-s3-headless-sim	PASS	Built and flashed for live headless simulation on the same physical controller.
waveshare_esp32_s3_rs485_can	PASS	Real headless target build still passes; behavior matched by headless simulation feature flags.
remote-crowpanel-21 / remote-crowpanel-128	PASS	Encoder remote builds previously passed with softened menu encoder path.

Live Connection Matrix Results

Both passes used the same physical controller; headless mode was simulated by disabling local UI features in board capabilities.

Case	Expected	Verification	Result
Full-size factory reset	No saved SSID, no saved peer.	Remote reset network + peer clear; controller reset network. Remote discovered controller from scratch.	PASS live
Full-size fresh discovery	ESP-NOW link first, Wi-Fi off.	Remote locked controller and stayed online with remote Wi-Fi inactive and staSaved=false.	PASS live
Full-size command matrix	Remote/controller sync in both directions.	RPM, weight, menu writes, mode toggles, and soak loop completed over ESP-NOW.	PASS live
Headless-sim factory reset	No saved SSID, no saved peer.	Remote reset network + peer clear; simulated headless controller reset network and remained recoverable.	PASS live
Headless-sim fresh discovery	Remote must find controller without Wi-Fi credentials.	Remote locked dual FE:01:2C:D6:38:BC, requested ESP-NOW-only, then settled on FC:01:2C:D6:38:BC.	PASS live
Headless-sim OFF guard	OFF must be refused when local UI is disabled.	API rejected radio OFF with remoteRecoveryRequired=true; controller stayed recoverable.	PASS live
Headless-sim command matrix	ESP-NOW remains primary under no-info setup.	RPM, weight, menu writes, mode toggles, and soak loop completed over ESP-NOW.	PASS live
WebSocket in no-info mode	Do not require Wi-Fi provisioning.	Skipped by design: a factory-fresh remote has no SSID and should not depend on WebSocket to recover.	N/A

Detailed Test Cases: Full-Size Controller

Normal Adafruit Feather controller build with local screen, buttons, encoder, and encoder button enabled.

ID	Case	Stimulus	Expected	Observed
F1	Factory/no-info reset	Remote reset_network + peer clear. Controller reset_network + idle. Controller radio set to WEB+ESP.	No saved home SSID on either recovery path and no pre-locked ESP-NOW peer.	PASS: remote staSaved=false; controller stayed recoverable on ESP-NOW channel 1.
F2	Fresh ESP-NOW discovery	Remote set to transport espnow after peer clear.	Remote should find the controller without Wi-Fi credentials, lock the peer, and report linkOnline.	PASS: remote locked FC:01:2C:D6:38:BC and linkOnline=true.
F3	Low-resource steady state	Let controller settle after discovery.	Once ESP-NOW is effective, Wi-Fi should not be required by the remote.	PASS: remote Wi-Fi inactive, staSaved=false, ESP-NOW online.
F4	Menu pull	Remote raw GET_MENU=4.	Remote can request and mirror controller menu data over ESP-NOW.	PASS: remote menu 4 data became visible; menu item check succeeded.
F5	Controller to remote RPM	Controller api setrpm 650.	Remote status should update controller.setRPM and manualRPM.	PASS: remote observed 650 RPM.
F6	Remote to controller RPM	Remote api setrpm 775.	Controller setRPM should update and remote should reflect it back.	PASS: controller and remote both observed 775 RPM.
F7	Weight sync both ways	Controller setweight 19000, then remote setweight 21000.	Both sides should converge on the latest target weight in mg.	PASS: controller and remote target weight values matched.
F8	Menu writes	Remote changed sleep timer, purge auto, and GBW RPM menu values.	Controller menu and remote menu mirrors should both converge.	PASS: all three menu writes were verified on controller and remote.
F9	State transitions	Remote action idle_gbw, then action idle.	Controller state should follow remote commands and remote status should mirror state.	PASS: IDLE_GBW then IDLE were both observed.
F10	ESP-NOW soak	10 iterations of remote RPM writes plus controller weight writes.	Link should remain ESP-NOW and linkOnline through repeated updates.	PASS: no link loss during soak.
F11	WebSocket under no-info	Skipped in factory/no-info run.	Fresh devices have no saved Wi-Fi; recovery must not depend on WebSocket.	N/A by design for this matrix.

Detailed Test Cases: Headless-Sim Controller

Same physical controller flashed with controller_box_headless_sim, reporting no local screen/buttons/encoder.

ID	Case	Stimulus	Expected	Observed
H1	Headless feature profile	Flash controller-universal-s3-headless-sim.	Controller target should report headless simulation so recovery rules treat local UI as unavailable.	PASS: target controller_box_headless_sim during test pass.
H2	Factory/no-info reset	Remote reset_network + peer clear. Controller reset_network + idle. Controller radio set to WEB+ESP.	Controller must remain remotely recoverable because there is no local UI fallback.	PASS: controller remained recoverable and remote had no saved Wi-Fi.
H3	Fresh ESP-NOW discovery	Remote set to transport espnow after peer clear.	Remote should find the headless controller without any saved Wi-Fi and lock the peer.	PASS: remote discovered the controller and came online over ESP-NOW.
H4	Dual to ESP-NOW settle	Controller moved from recovery/dual advertisement toward ESP-NOW-only steady state.	Remote may see the temporary FE:01:2C:D6:38:BC dual MAC, then settle on FC:01:2C:D6:38:BC.	PASS: final locked peer was FC:01:2C:D6:38:BC and linkOnline=true.
H5	OFF guard	Controller api set 1 4 2 while remoteRecoveryRequired=true.	Radio OFF must be rejected so a headless controller is not stranded.	PASS: command returned ok=false with headless rejection; desired mode stayed non-OFF.
H6	Menu pull	Remote raw GET_MENU=4.	Remote can request controller menu data over ESP-NOW after headless recovery.	PASS: remote menu data check succeeded.
H7	RPM and weight sync	Same controller-to-remote and remote-to-controller RPM/weight checks as full-size.	Both sides should converge without WebSocket and without local UI.	PASS: setRPM and setWeight paths worked both directions.
H8	Menu writes	Remote changed sleep timer, purge auto, and GBW RPM menu values.	Controller and remote menu mirrors should both converge in headless mode.	PASS: all menu writes were verified.
H9	State transitions	Remote action idle_gbw, then action idle.	Headless controller should still follow state commands over ESP-NOW.	PASS: IDLE_GBW then IDLE were both observed.
H10	ESP-NOW soak	10 iterations of remote RPM writes plus controller weight writes.	Link should remain ESP-NOW and linkOnline through repeated updates.	PASS: no persistent link loss during soak.
H11	Restore bench unit	Reflash controller-universal-s3 after headless-sim pass.	Physical controller should return to normal full-size firmware.	PASS: target controller_box, board Adafruit Feather Controller, remoteRecoveryRequired=false.

Recovery Paths And Issue Notes

This page records what should happen when a case hits a fault or ambiguous transition.

Scenario	Trigger	Recovery Path	Bench Result
Factory no saved Wi-Fi / no saved peer	Controller starts WEB+ESP recovery. Remote starts ESP-NOW scan with no locked peer.	Remote discovers controller, locks peer, establishes ESP-NOW, and final state can drop remote Wi-Fi use.	Verified in full-size and headless-sim passes.
Remote does not find ESP-NOW controller	Remote has no established ESP-NOW controller. If Wi-Fi is configured, it may try Wi-Fi discovery; with no Wi-Fi it stays idle and waits for scan/setup.	Controller must not turn itself into an unrecoverable OFF state while waiting.	Factory/no-info run verified the no-Wi-Fi branch does not require WebSocket.
Headless controller receives radio OFF request	API/menu request tries to set radio mode OFF while remoteRecoveryRequired=true.	Reject OFF, keep desired mode recoverable, and keep ESP-NOW/recovery path alive.	Verified: api set 1 4 2 returned ok=false and desired mode stayed non-OFF.
Full-size controller receives radio OFF request	Local screen/buttons/encoder are present.	OFF may remain allowed because the user has a local recovery path on the controller.	Not part of no-strand headless failure path; guard intentionally only applies when local UI is disabled.
Dual MAC to ESP-NOW MAC transition	During headless recovery, remote can first observe FE:01:2C:D6:38:BC and then settle to FC:01:2C:D6:38:BC.	A transient offline sample should trigger a bounded relock wait rather than marking the controller stranded.	Observed once in early headless load probe; harness updated to wait up to 10s and final rerun passed.
Earlier WebSocket failure	One obsolete run mixed a controller with no saved STA and a remote with a saved home SSID.	Reset both devices for a real factory/no-info test and skip WebSocket because no saved Wi-Fi exists.	Resolved by factory reset path; not counted as a product failure.

Load / Hogging Check

Serial/API round-trip timings during the live matrix. Max threshold used by the harness: 1500 ms.

Probe	Side	N	Avg	Max	Result
Full-size fresh ESP-NOW discovery	Remote	12	32.1 ms	130.9 ms	PASS
Full-size fresh ESP-NOW discovery	Controller	12	141.1 ms	325.5 ms	PASS
Full-size ESP-NOW command suite	Remote	12	17.5 ms	133.7 ms	PASS
Full-size ESP-NOW command suite	Controller	12	119.7 ms	293.4 ms	PASS
Headless-sim fresh ESP-NOW discovery	Remote	12	24.9 ms	110.8 ms	PASS
Headless-sim fresh ESP-NOW discovery	Controller	12	97.3 ms	165.4 ms	PASS
Headless-sim OFF guard	Remote	12	9.4 ms	40.5 ms	PASS
Headless-sim OFF guard	Controller	12	93.2 ms	135.6 ms	PASS
Headless-sim ESP-NOW command suite	Remote	12	21.3 ms	107.7 ms	PASS
Headless-sim ESP-NOW command suite	Controller	12	81.6 ms	151.0 ms	PASS

Overall Pass	Side	Count	Avg	P95	Max
Full-size no-info pass	Remote	114	18.9 ms	113.5 ms	133.7 ms
Full-size no-info pass	Controller	78	145.5 ms	335.7 ms	343.1 ms
Headless-sim no-info pass	Remote	141	13.8 ms	88.5 ms	111.8 ms
Headless-sim no-info pass	Controller	83	101.0 ms	222.0 ms	290.0 ms

Result:

1. No load probe exceeded the 1500 ms guard threshold; no periodic hogging was observed in this matrix.
2. Factory/no-info WebSocket was intentionally skipped; recovery must not depend on saved Wi-Fi.
3. Follow-up UI notes captured separately: remote centering across screens and clock temperature suffix.